KU LEUVEN

ARENBERG DOCTORAL SCHOOL
Faculty of Engineering Technology

Dynamic reality modeling for AECO applications

Jelle Vermandere

Dissertation presented in partial fulfillment
of the requirements for the degree of
Doctor of Engineering Technology (PhD)

October 2026

Supervisors:
Prof. dr. ir. M. Vergauwen
Prof. dr. ing. M. Bassier

DYNAMIC REALITY MODELING FOR AECO APPLICATIONS

Jelle VERMANDERE

Supervisors:
Prof. dr. ir. M. Vergauwen
Prof. dr. ing. M. Bassier

Members of the
Examination Committee:
Prof. dr. ir.???, chair
Prof. dr. ir. C. Cannaerts
Prof. dr. Adalberto L. Simeone
Kristof Vandenbogaerde
(Besix)

Dissertation presented in partial fulfillment of the requirements for the degree of Doctor of Engineering Technology (PhD)

October 2026

© 2026 Jelle Vermandere
Uitgegeven in eigen beheer, Jelle Vermandere, Gent (Belgium)

Alle rechten voorbehouden. Niets uit deze uitgave mag worden vermenigvuldigd en/of openbaar gemaakt worden door middel van druk, fotokopie, microfilm, elektronisch of op welke andere wijze ook zonder voorafgaande schriftelijke toestemming van de uitgever.

All rights reserved. No part of the publication may be reproduced in any form by print, photoprint, microfilm, electronic or any other means without written permission from the publisher.

Contents

Contents i

List of Abbreviations vi

1 Introduction 1
1.1 General context 1
1.2 Problem statement 4
1.3 Research aim 5
1.4 Thesis Outline 6

2 Scene Dynamification Background and Terminology 9
2.1 Introduction 9
2.2 3D Representations 10
2.2.1 Explicit Representations 10
2.2.2 Implicit Representations 11
2.2.3 Representational Choice in This Thesis 13
2.3 Dynamic Reality Model Requirements 13
2.3.1 Inputs 13
2.3.2 Output Requirements 14
2.4 Pipeline Overview 15
2.4.1 The Scene–Object Completion Duality 15
2.4.2 Pipeline Stages 16
2.5 Existing Approaches to Scene Dynamification 18
2.6 Scope and Assumptions 19

3 Data Acquisition 21
3.1 Introduction 21
3.2 Related Work: Capture Technologies and Datasets 22
3.2.1 Capture Technologies 22
3.2.2 Limitations of Real and Synthetic Data 23
3.3 Real-World Data Capture 24

3.3.1 Laser Scanner Captures 24
3.3.2 XR Device Captures 26
3.3.3 Public Real-World Datasets 26
3.3.4 Evaluation and Shortcomings of Real-World Data . . . 28
3.4 Synthetic Data Creation 29
3.4.1 Related Work: Synthetic Datasets and Virtual Scanners 29
3.4.2 Methodology: The Virtual Scanner 30
3.4.3 Procedural Scene Generation 32
3.4.4 The V-Scan Dataset 33
3.5 Conclusion 34

4 Alignment And Updating 37

5 Scene Segmentation 39
5.1 Introduction 39
5.2 Related work 39
5.3 Methodology 39
5.3.1 Object detection 39
5.3.2 Environment segmentation 39
5.4 Experiments 39
5.5 Conclusion 39

6 Scene Completion 41
6.1 Introduction 41
6.2 Related work 41
6.3 Methodology 41
6.3.1 Geometry reconstruction 41
6.3.2 Color inpainting 41
6.4 Experiments 41
6.5 Conclusion 41

7 Object Completion 43
7.1 Introduction 43
7.2 Related work 43
7.3 Methodology 43
7.3.1 Geometry generation 43
7.3.2 Texture generation 43
7.4 Experiments 43
7.5 Conclusion 43

8 Scene Dynamification 45
8.1 Introduction 46
8.2 Related work 46

8.3 Object Dynamification 46
8.3.1 Part segmentation 46
8.3.2 Scaling zone definition 46
8.3.3 Value interpolation 46
8.3.4 Value repetition 46
8.4 Scene Reconstruction 46
8.5 Experiments 46
8.6 Conclusion 46

9 Conclusion and future works 47
9.1 Conclusion 47
9.2 Future works 47

10 Valorisation 49
10.1 Introduction 49
10.2 Proof-of-concept applications 49
10.2.1 Digital upcycling 49
10.2.2 Furniture Removal 49
10.3 Valorization potential and market position 49
10.3.1 Time savings analysis 49
10.4 Conclusion 49

Bibliography 51

Curriculum Vitae 57

Publications 59

List of Abbreviations

AECO Architecture, Engineering, Construction and Operations. 1

BIM Building Information Model. 1, 12

CSDF Coloured Signed Distance Field — an extension of the SDF representation in which each voxel additionally stores per-voxel colour information, enabling joint geometry and appearance manipulation. 12

DRM Dynamic Reality Model. 2, 3, 5–7, 9, 12–15, 17, 19

IMU Inertial Measurement Unit — a sensor that measures acceleration and angular velocity to support motion tracking and pose estimation. 22

MMS Mobile Mapping System — a laser scanning platform mounted on a moving carrier, enabling continuous data capture along a trajectory. 22

NeRF Neural Radiance Field — a learned continuous function mapping 3D coordinates and viewing directions to colour and volume density, enabling novel view synthesis. 12

RGB-D Red-Green-Blue plus Depth — a data modality combining a colour image with a registered per-pixel depth map. 26

SDF Signed Distance Function. 3, 12

SfM Structure from Motion — a photogrammetric technique for reconstructing 3D structure from a set of overlapping 2D images. 23

SLAM Simultaneous Localisation and Mapping — an algorithm for concurrently estimating a sensor's position and building a map of the surrounding environment. 22

TLS Terrestrial Laser Scanner — a static, tripod-mounted laser scanning device that captures high-precision point clouds from a fixed position. 22

ToF Time-of-Flight — a depth sensing technology that measures distance by timing the return of emitted light pulses. 23

UDF Unsigned Distance Field — a scalar field encoding the absolute (unsigned) distance from any point in space to the nearest surface, applicable to open or non-watertight geometry. 12

UV UV coordinates — two-dimensional texture coordinates assigned to mesh vertices that define the mapping between the 3D surface and a 2D texture image. 10

XR Extended Reality. 3, 10, 13

Chapter 1

Introduction

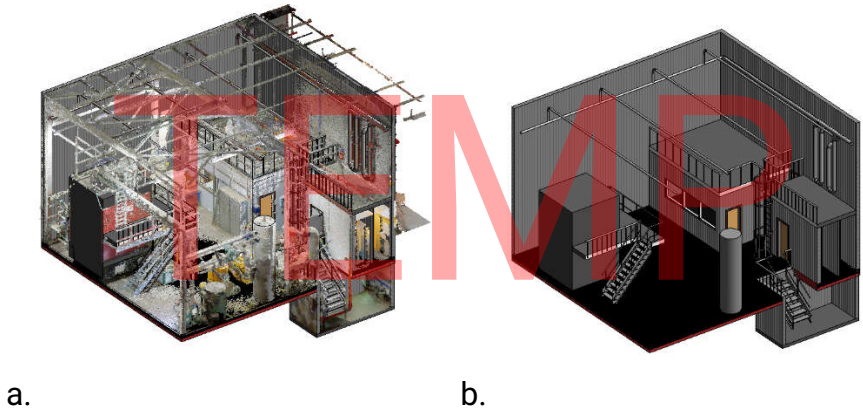
1.1 General context

The Architecture, Engineering, Construction and Operations (AECO) industry is undergoing a rapid digital transformation, driven by advances in reality capture technologies such as laser scanning and photogrammetry. These technologies enable the efficient acquisition of high-resolution representations of the physical environment, forming a foundation for data-driven workflows and digital twins.

Two complementary paradigms currently dominate the representation of the built environment. On the one hand, reality capture-based models provide highly detailed and visually accurate reconstructions of real-world scenes, making them well suited for documentation, inspection and visualization. On the other hand, Building Information Modelling (BIM) represents environments as structured collections of semantic, parametric objects, enabling efficient querying, modification and simulation.

Each paradigm offers distinct advantages, but also exhibits inherent limitations. Reality capture models preserve geometric and visual fidelity, yet remain static, incomplete and unstructured, lacking explicit object decomposition and semantic relationships. In contrast, BIM models provide a dynamic and structured representation of buildings, but often lack geometric detail and visual realism, as they are typically created during the design phase and do not fully reflect as-built conditions. (Figure 1.1)

This contrast highlights a fundamental limitation in current digital modelling



TEMP

Figure 1.1: Comparison between reality capture and Building Information Modelling approaches.

workflows: no single representation simultaneously provides realism, completeness and interactivity. Bridging this gap requires combining the strengths of both paradigms into a unified representation.

To address this challenge, this research introduces a novel representation, referred to as a *Dynamic Reality Model* (DRM). The DRM represents the environment as a collection of individual, manipulable objects, while preserving the geometric and visual fidelity of the captured scene. In doing so, it combines the structured and dynamic nature of BIM with the highly accurate and visually detailed reality capture data.

The resulting representation enables a wide range of applications that require both accuracy and flexibility. In renovation and refurbishment workflows, designers can evaluate design alternatives directly within the as-is environment, supporting tasks such as layout optimisation and feasibility analysis without requiring full manual reconstruction. In facility management and reorganisation scenarios, objects such as furniture or equipment can be repositioned, replaced or removed, enabling efficient space planning while maintaining consistency with real-world constraints. Furthermore, the combination of visual fidelity and object-level interactivity enables the use of real-world environments in interactive applications such as simulation, training and gaming.

Despite this potential, a fundamental barrier remains in transforming reality capture data into such dynamic representations. This barrier stems from the

combined effects of occlusions and the unstructured nature of the captured data. Scan-based representations only capture visible surfaces from limited viewpoints, leaving significant portions of the environment unobserved due to occlusions caused by clutter, other objects or restricted sensor access. At the same time, the data lacks an explicit structural organisation, representing the scene as a dense collection of points or surfaces without clear object boundaries.

While recent learning-based approaches attempt to address the issue of missing geometry using implicit representations such as SDFs [19] or occupancy fields [10], and scene-level methods infer unseen regions [11], these approaches primarily focus on geometric completion. They do not address the combined challenge of reconstructing missing data while simultaneously introducing the structured, object-level representation required for interaction.

In addition to these structural challenges, built environments are inherently dynamic, continuously evolving over time due to usage, modifications and maintenance activities. As a result, any DRM that aims to accurately represent the physical environment must be updated regularly to remain relevant.

Traditionally, such updates rely on high-end sensing technologies, such as terrestrial laser scanning, which provide accurate and detailed observations but are time-consuming and costly to acquire. This limits the frequency at which complete datasets can be captured. In contrast, recent advances in mobile and XR-based capture devices enable fast and accessible data acquisition, allowing smaller updates to be recorded more frequently, albeit at lower accuracy and completeness.

This creates an opportunity to complement high-quality baseline datasets with incremental updates from lower-end devices. However, it also introduces new challenges, as the resulting data is heterogeneous in terms of accuracy, density and coverage. Integrating these observations into a consistent and up-to-date representation requires careful handling of temporal differences, data quality and spatial alignment [4, 25].

As a result, despite significant advances in data acquisition and reconstruction, there is currently no unified framework that transforms partial, heterogeneous scan data into complete, structured and dynamic scene representations. Existing methods address individual aspects of this problem, but fail to integrate them into a coherent and scalable pipeline.

1.2 Problem statement

Transforming partially scanned indoor environments into complete, structured and dynamic representations remains an open challenge in current 3D reconstruction workflows. This challenge is inherently multi-stage, requiring the integration of heterogeneous data, the recovery of missing geometry and the transformation of static reconstructions into interactive environments.

A first challenge lies in the alignment and updating of heterogeneous data. Different sensing technologies produce data with varying levels of accuracy, density and noise, resulting in inconsistent representations of the same environment. Aligning these data sources into a common reference frame is particularly difficult in indoor settings, where global positioning systems are unavailable. In addition, environments evolve over time, requiring new partial observations to be integrated into existing datasets. Efficiently updating scenes without reprocessing the entire dataset remains a significant challenge, requiring robust methods for both registration and incremental integration.

A second major challenge is the incompleteness of reality capture data. Scan-based devices record only what is visible from their measurement positions, leaving occluded regions unobserved. This affects both the background scene and individual objects, producing data with missing geometry at multiple scales. Effective completion additionally requires the reliable separation of individual objects from their surroundings, which is particularly difficult in cluttered environments where object boundaries are ambiguous and partially hidden.

A deeper difficulty is that scene-level and object-level completion are typically treated as separate problems, each with limitations that the other cannot resolve. Scene-level completion methods [16, 8, 28, 12, 13, 5, 23, 20] treat the environment as a monolithic volume and infer missing geometry globally. While this recovers regions outside the scanner's field of view, it cannot resolve occlusions between objects that are in contact: two chairs pushed together, a cabinet against a wall, or stacked equipment all present hidden surfaces that a global scene model cannot distinguish from empty space. Object-level completion methods [19, 10, 22, 9], by contrast, operate on objects already isolated from the scene, discarding the very environmental context — surrounding walls, floors and neighbouring objects — that could inform the plausible extent and boundaries of the missing geometry. Resolving this mutual dependency requires a framework in which scene and object completion are coupled through shared geometric constraints rather than treated in isolation.

Even when complete geometric reconstructions of both the scene and individual objects are available, transforming them into a dynamic and interactive

environment remains a non-trivial problem. This requires their analysis and structuring to support higher-level interactions. Objects must be represented in a way that enables operations such as repositioning, replacement or modification, which goes beyond purely geometric reconstruction.

A final challenge lies in the evaluation of reconstruction and completion methods. Reliable ground-truth data for occluded regions is rarely available in real-world datasets, making quantitative evaluation difficult. The absence of ground truth for hidden surfaces means that existing benchmarks cannot directly assess the quality of completed geometry in the regions that matter most. Synthetic datasets offer an alternative, but must accurately replicate real-world scanning conditions — including noise, occlusion patterns and sensor-specific characteristics — to produce evaluation results that transfer to practice.

In summary, transforming partial scan data into dynamic scene representations requires addressing challenges in data alignment and updating, geometry completion at both scene and object level, and scene dynamification. These challenges are interdependent and must be solved within a unified and structured framework.

1.3 Research aim

The primary objective of this research is to develop a modular and adaptable methodology for transforming partially scanned indoor environments into complete, structured and dynamic representations, referred to as DRMs. This objective addresses the lack of a unified pipeline identified in the previous section, where current approaches treat alignment, completion and interaction as separate problems rather than components of a coherent workflow.

Instead of relying on a monolithic end-to-end solution, the proposed methodology is structured as a sequence of interconnected modules. Each module performs a specific task, producing an output that serves as input for the next stage. This modular design enables flexibility in handling heterogeneous and incomplete data, while allowing intermediate validation and refinement when required.

The pipeline consists of the following stages, as outlined in Figure 1.2:

- **Scene alignment and updating**, integrating heterogeneous and multi-temporal data into a consistent representation while supporting incremental updates.
- **Scene segmentation and object isolation**, decomposing unstructured scan data into meaningful object-level components.

- **Scene completion**, reconstructing missing structural elements caused by occlusions.
- **Object completion**, recovering partially observed objects while preserving their contextual relation to the scene.
- **Scene dynamification**, transforming static reconstructions into structured, interactive representations that support object-level manipulation.

The conceptual role of each stage and the terminology associated with it are defined in Chapter 2, which establishes the representational and methodological foundation for the pipeline.

A key contribution of this work is the development of a DRM, which combines the geometric and visual fidelity of reality capture data with the structured and interactive nature of object-based models. This enables objects to be manipulated, repositioned and analysed within a representation that remains consistent with the physical environment.

To support evaluation, this research introduces a synthetic dataset generation framework that produces partial scans paired with complete ground-truth geometry. Because real-world datasets cannot provide ground truth for occluded regions — the very regions that completion methods must reconstruct — such a synthetic dataset is essential for quantitative assessment under controlled conditions that reflect real-world scanning characteristics.

The scope of this research is limited to single-room indoor environments containing furniture or industrial objects. This constraint enables a focused investigation of densely furnished scenes, where occlusions, object interactions and data heterogeneity are significant but manageable.

In summary, this research establishes a modular pipeline that integrates alignment, segmentation, completion and dynamification into a coherent workflow. By combining the strengths of reality capture and structured representations, the proposed methodology enables the creation of dynamic models from partial scan data.

1.4 Thesis Outline

This thesis follows the logical progression from data acquisition to the generation of dynamic scene representations. Each chapter builds upon the previous one forming a coherent pipeline from raw scan data to DRMs.

Chapter 2 introduces the background and terminology related to scene dynamification and 3D representations. It defines key concepts such as dynamic scenes, object-level representations and completion, positions the proposed DRM within the broader context of existing modelling paradigms, and explains the conceptual role of each pipeline stage.

Chapter 3 focuses on data acquisition, covering both real-world scanning and synthetic dataset generation. It discusses the characteristics and limitations of different sensing technologies, including occlusions, noise and coverage, and introduces the V-Scan synthetic dataset designed to enable controlled evaluation of reconstruction methods.

Chapter 4 addresses the alignment and updating of multi-source and multi-temporal data. It presents methods for integrating heterogeneous datasets into a consistent representation and explores strategies for incrementally updating scenes as new observations become available.

Chapter 5 focuses on object detection and occlusion analysis. It introduces methods for decomposing unstructured scan data into individual objects and for estimating occluded regions at both object and scene level, providing the basis for subsequent completion steps.

Chapter 6 presents the scene completion methodology, focusing on the reconstruction of structural elements of the environment. It describes how incomplete observations are converted into a consistent geometric and textured representation of the scene.

Chapter 7 explores object completion techniques, comparing different approaches for reconstructing partially observed objects. The chapter evaluates their performance and discusses their integration within the overall pipeline.

Chapter 8 introduces the dynamification of completed scenes. It describes how static reconstructions are transformed into structured, interactive representations, enabling object-level manipulation and interaction within the environment.

Chapter 9 concludes the thesis by summarizing the main contributions and discussing the limitations of the proposed methodology, as well as directions for future research.

Chapter 10 explores the valorization potential of the research, highlighting applications in areas such as renovation, facility management and interactive environments, and discussing opportunities for real-world deployment.

Through this structured approach, the thesis reflects the full transformation pipeline from partial scan data to dynamic and interactive scene representations.

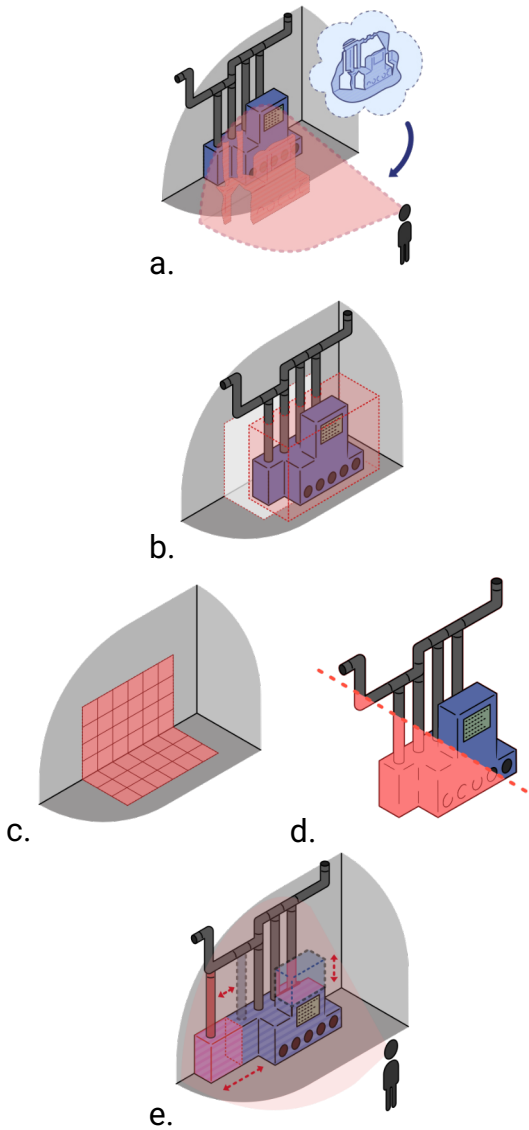


Figure 1.2: Overview of the proposed modular pipeline for transforming partial scans into dynamic reality models.

Chapter 2

Scene Dynamification Background and Terminology

2.1 Introduction

The previous chapter introduced the DRM as a representation that combines the geometric and visual fidelity of reality capture with the structured, interactive nature of object-based models, and identified the pipeline of stages required to produce it. Before presenting each stage in detail, this chapter establishes the conceptual and terminological foundation on which the remainder of this thesis rests.

Building on the representational definition of Chapter 1, we now define *scene dynamification* as the *process* of producing a DRM: transforming a static, unstructured 3D reconstruction into a structured collection of individually manipulable objects, embedded within a geometrically and visually complete environment. Rather than treating a scanned scene as a monolithic surface, dynamification introduces an explicit object-level decomposition that enables operations such as repositioning, replacement, scaling and removal, while preserving consistency with the captured geometry and appearance.

This chapter defines the scope and requirements of dynamification and positions it relative to existing representations. Section 2.2 reviews the landscape of 3D representations and discusses their suitability as inputs and outputs of the proposed pipeline. Section 2.3 formalises the requirements of a DRM in terms of its inputs, outputs and structural properties, including a typology of the

occlusion categories that the pipeline must address. Section 2.4 provides the conceptual description of the pipeline stages and explains the design rationale for treating scene and object completion in parallel. Section 2.5 surveys existing work that most directly addresses scene dynamification as a unified problem. Section 2.6 concludes with a summary of the scope and assumptions adopted throughout this thesis.

2.2 3D Representations

A prerequisite for understanding scene dynamification is a clear picture of the representational choices available for encoding 3D geometry and appearance. These choices have direct consequences for what operations are possible, what completion methods can be applied, and how the final dynamic scene can be consumed by downstream applications. Representations can be broadly divided into *explicit* and *implicit* categories, each with distinct trade-offs in terms of resolution, editability, storage efficiency and support for texture.

2.2.1 Explicit Representations

Explicit representations directly encode the geometry of a surface or volume as a discrete set of geometric primitives. They are generally well suited for visualization and downstream processing, as the geometry is directly accessible without the need for additional decoding steps.

Point clouds. A point cloud is an unordered set of 3D coordinates, typically accompanied by surface normals and colour values per point. Point clouds are the natural output of laser scanning and depth-sensing devices, and they form the primary raw input modality for the pipeline developed in this thesis. Their main advantages are simplicity, scalability and direct correspondence to physical measurements. However, they lack surface connectivity, which complicates downstream operations such as rendering, meshing and UV mapping. Colour is typically stored per point as an RGB triplet, either captured directly by the sensor or assigned through projection from a co-registered image [3]. The density and noise characteristics of a point cloud depend strongly on the sensing technology used, ranging from sub-millimetre precision in high-end terrestrial laser scanners to centimetre-level accuracy in mobile and XR-based devices [26].

Meshes. A polygonal mesh represents a surface as a set of vertices connected by edges and faces, most commonly triangles. Meshes provide explicit surface connectivity and are the standard representation for rendering, simulation and geometric processing. They support a rich set of operations including Boolean combinations, normal computation and UV parameterisation for texture mapping.

Texture in meshes is typically stored as a 2D image, indexed via UV coordinates assigned to each vertex. Each face samples its colour from this texture image using barycentric interpolation of the UV coordinates of its three vertices [15]. This decoupling between geometry and appearance allows high-resolution textures to be stored efficiently, but introduces a non-trivial mapping between the 3D surface and the 2D texture space, which can complicate texture inpainting and completion [24].

Meshes obtained from indoor scans are typically the result of a reconstruction pipeline applied to point clouds or depth images, and may therefore contain artefacts such as holes, non-manifold geometry or incorrect connectivity. Processing such meshes requires dedicated repair steps before further operations can be applied [**?**].

3D Gaussian splatting. A more recent explicit representation is 3D Gaussian splatting [**?**], in which the scene is represented as a set of anisotropic 3D Gaussians, each parameterised by a position, covariance, opacity and spherical harmonic coefficients encoding view-dependent colour. This representation has gained considerable attention for its ability to achieve real-time novel view synthesis at high visual fidelity. Each Gaussian implicitly encodes both geometry and appearance, analogous to coloured point clouds but with a richer per-element appearance model. While segmentation of Gaussian splat scenes into individual objects is an active area of research [1], mature tools for object-level geometric editing, completion and part-aware deformation of Gaussian representations do not yet exist. Furthermore, the lack of an explicit, watertight surface definition makes completion and UV-based texture operations difficult to apply. For these reasons, 3D Gaussian splatting is not adopted as an intermediate or output representation in this thesis.

2.2.2 Implicit Representations

Implicit representations encode geometry indirectly, typically as a continuous scalar field over 3D space whose zero level set defines the object surface. They offer smooth, resolution-independent surface descriptions and are particularly

well suited as intermediate representations in learning-based completion and generation methods.

Signed Distance Functions. A SDF assigns to each point in space a scalar value representing the signed distance to the nearest surface, with negative values inside the object and positive values outside. The surface itself is recovered as the zero level set of this function. When discretised over a regular voxel grid, an SDF provides a fixed-size, differentiable representation that is compatible with convolutional neural network architectures [19]. This makes SDFs a popular input and output format for object completion networks. However, conventional SDFs require the underlying geometry to be watertight, a condition rarely satisfied by raw scan data. UDFs, which encode only the absolute distance to the surface, provide a more general alternative applicable to open or incomplete meshes [27].

For the dynamification step, SDFs can be extended to CSDFs by additionally storing per-voxel colour information, enabling joint geometry and appearance manipulation [**?**].

Parametric and semantic models. BIM models represent buildings as structured collections of parametric objects with associated semantic properties. Each element, such as a wall, door or piece of furniture, is described by its type, dimensions and relationships to other elements, rather than by explicit surface geometry. This structured representation supports efficient querying, simulation and modification, but typically lacks the geometric and visual detail of captured data [2]. As discussed in Chapter 1, the gap between BIM and reality capture representations motivates the development of the DRM.

Neural implicit representations. NeRFs [18] and related approaches represent scenes as learned continuous functions mapping 3D coordinates and viewing directions to colour and volume density. These representations achieve compelling novel view synthesis quality but are generally not editable at the object level, making them unsuitable as output representations for dynamification. Recent methods attempt to decompose NeRF scenes into individual objects [1], but this remains an active research challenge and is not further pursued in this thesis.

2.2.3 Representational Choice in This Thesis

Given the above considerations, the pipeline developed in this thesis operates primarily with coloured point clouds and textured meshes as the main data types, with SDFs and UDFs as intermediate representations in completion stages. Point clouds serve as the natural output of the capture devices used and as the input to detection and completion modules. Textured meshes serve as the output of the scene completion stage, providing a compact, renderable and editable surface representation. SDFs are used internally within object completion networks, bridging the gap between the irregular scan data and the fixed-size inputs required by learning-based models.

This choice reflects a deliberate balance between data fidelity, geometric expressiveness and compatibility with standard tools and processing pipelines. BIM and neural representations are explicitly excluded from the pipeline output, as they either lack visual fidelity or do not support the object-level editing operations required for dynamification.

2.3 Dynamic Reality Model Requirements

Having established the representational landscape, we can now formalise what a DRM must provide and what inputs it operates on. This section defines the scope and structural requirements of the proposed representation.

2.3.1 Inputs

The pipeline takes as input heterogeneous, multi-source data captured from real indoor environments. This data is inherently incomplete, as no single scan captures a scene in its entirety. Two primary modalities are considered.

Point clouds. Coloured, unstructured point clouds form the geometric backbone of the input data. These may originate from high-end terrestrial laser scanners, which provide millimetre-level accuracy over large ranges, or from lower-end mobile and XR-based devices, which sacrifice accuracy for speed and accessibility [26]. As discussed in Chapter 1, no single scan captures a scene in its entirety: the point cloud covers only surfaces visible from the scanner positions, leaving occluded regions unobserved. Combining multiple scans from different positions increases coverage but introduces alignment challenges due to noise and drift [25].

Image data. Panoramic images or localised pinhole camera images provide complementary appearance information. Panoramic images, captured at the same positions as the scan data, enable dense colour assignment to the point cloud through angular projection. Localised images from mobile devices provide additional texture cues and can support 2D-based alignment and inpainting operations. The co-localisation of images within the same coordinate system as the point cloud is a prerequisite for cross-modal processing [4].

2.3.2 Output Requirements

The output of the pipeline is a DRM: a structured, complete and interactive representation of the indoor environment. This representation must satisfy the following requirements.

Geometric completeness. Both the structural elements of the scene and the individual objects within it must be geometrically complete. Occluded or unobserved regions must be reconstructed so that the resulting representation does not contain holes or missing surfaces. This applies at two levels: the background scene, consisting of walls, floors and ceilings, and the individual foreground objects, such as furniture or equipment.

It is useful to distinguish three categories of occlusion that the pipeline must address. *Sensor occlusion* refers to regions that fall outside the scanner's line of sight entirely, such as the back of a room not covered by any scan position. *Inter-object occlusion* refers to surfaces of one object that are hidden behind another, such as the rear face of a shelf visible only if the objects in front of it are removed. *Contact occlusion* refers to surfaces hidden because two objects are physically touching, such as the base of a chair leg resting on a floor or the back of a cabinet flush against a wall. Each category requires a different treatment: sensor occlusions can often be resolved by additional scan positions or scene-level inference, inter-object occlusions require object-level reasoning, and contact occlusions require knowledge of both the object geometry and the surrounding scene structure. The occlusion analysis developed in Chapter 5 explicitly estimates all three types and passes this information to the scene and object completion stages.

Visual completeness. The completed geometry must be accompanied by plausible texture. Missing texture regions that arise from occlusions or incomplete image coverage must be inpainted in a manner that is consistent

with the surrounding appearance. The resulting textured representation must be visually coherent across the entire scene.

Object-level decomposition. Individual objects must be represented as separate entities within the scene, rather than as part of a monolithic surface. This decomposition is essential for enabling object-level manipulation. Each object must be isolatable from the scene without introducing inconsistencies in the surrounding environment.

Part awareness and scalability. Part awareness in this pipeline is not a post-hoc annotation step but emerges naturally from the object completion process. The material segmentation performed during texture completion in Chapter 7 assigns discrete material indices to distinct surface regions of each object. These indices are propagated to the completed geometry and serve directly as the input to the convex decomposition and scaling zone definition in Chapter 8. This coupling ensures that the part structure used for scaling is grounded in the actual material composition of the completed object rather than in a generic, class-specific template.

Manipulability. Objects within the DRM must support a set of basic manipulation operations: repositioning within the scene, removal from the scene, and scaling. These operations must be applicable without re-running the full pipeline and without introducing visual artefacts in the surrounding scene.

2.4 Pipeline Overview

The requirements defined above are addressed by a modular pipeline, each stage of which is developed in detail in subsequent chapters. This section provides the conceptual description of each stage and its role within the overall workflow, fulfilling the pointer made in Chapter 1. A key design decision running through the pipeline is the parallel treatment of scene and object completion, which is motivated here before the individual stages are described.

2.4.1 The Scene–Object Completion Duality

A central challenge in producing complete dynamic scenes is that scene-level and object-level completion are mutually dependent yet conventionally treated as

separate problems. Scene-level completion methods infer missing geometry by reasoning over the environment as a whole. This works well for regions that were simply outside the scanner's field of view, but fails at contact occlusions: surfaces hidden because two objects are physically touching cannot be distinguished from empty space by a global volumetric model, which has no concept of object identity or boundary.

Object-level completion methods operate on objects after they have been isolated from the scene. Isolation, however, strips away the environmental context that is most informative for completion. The floor surface constrains the bottom extent of a chair; the wall behind a cabinet defines its maximum depth; a neighbouring object provides an upper bound on where one object can end and another begin. A completion network that operates on a centred, decontextualised object mesh has none of this information available.

The proposed pipeline resolves this mutual dependency by decoupling detection from completion while keeping both coupled through a shared geometric analysis. Object detection produces bounding boxes that are refined using scene geometry — walls, floors and symmetry axes — before any completion is attempted. A voxel-based occlusion analysis then produces separate grids at the object level and scene level. These grids encode which regions are genuinely missing versus which were simply not observed, and they are passed simultaneously to the scene completion stage, which uses them to avoid reconstructing geometry in regions occupied by detected objects, and to the object completion stage, which uses them to define the voxels that the completion network must predict. The result is a pair of complementary reconstructions that are geometrically consistent with one another from the outset.

2.4.2 Pipeline Stages

Scene alignment and updating. Before any completion or decomposition can take place, multi-source and multi-temporal data must be brought into a common coordinate system. This stage handles the registration of point clouds and images from heterogeneous devices and integrates incremental updates from lower-end sensors into existing high-quality datasets. The output is a consistently georeferenced, multi-modal dataset representing the current state of the environment. This stage is addressed in Chapter 4.

Object detection and occlusion analysis. The aligned point cloud is decomposed into individual objects using 3D detection methods. Each detected object is associated with an oriented bounding box that is refined using symmetry

estimation and scene geometry constraints such as walls and floors. In parallel, a voxel-based occlusion analysis estimates the occluded regions of both individual objects and the full scene, distinguishing the three occlusion categories defined in Section 2.3.2. The resulting grids provide the shared geometric information that couples the scene and object completion stages. This stage is addressed in Chapter 5.

Scene completion. The structural elements of the environment, separated from the detected objects, are completed geometrically and texturally. Planar structures such as walls, floors and ceilings are reconstructed using plane detection and intersection-based geometry generation, guided by the scene-level occlusion grid to restrict reconstruction to genuinely missing regions and to avoid filling space occupied by detected objects. Texture is inpainted on the reconstructed geometry using the panoramic image data as reference. The output is a complete, textured mesh of the background scene. This stage is addressed in Chapter 6.

Object completion. Individual objects are completed independently, using both the partial point cloud and the co-registered image data as input. The object-level occlusion grid defines which voxels the completion network must predict, while the refined bounding box — constrained by the scene planes — defines the physical extent within which the completion is permitted. The completed objects are represented as textured meshes and carry part-index information derived from material segmentation, enabling the subsequent dynamification step. This stage is addressed in Chapter 7.

Scene dynamification. The completed objects are transformed into manipulable entities by applying part-aware scaling and decomposition. Each object's internal structure is analysed and divided into approximately convex components that serve as the basis for scaling zones, using the material indices produced during object completion as structural priors. The completed scene and its objects are then assembled into a DRM in which each object can be independently repositioned, scaled or removed. This stage is addressed in Chapter 8.

2.5 Existing Approaches to Scene Dynamification

Although the individual components of the proposed pipeline, such as object detection, completion and inpainting, have been studied extensively in isolation, the specific problem of transforming a partial scan into a fully dynamic and editable scene representation has received comparatively little attention.

The closest relevant body of work includes methods for indoor scene understanding that decompose a scan into structural and object components. Li et al. [17] segment indoor point clouds into planar structural elements and detected objects, reconstruct the structural geometry via plane fitting, and replace detected objects with matched CAD models retrieved from a database. This approach produces a lightweight and semantically structured scene, but is limited to objects that have a close match in a predefined database and does not address the visual completeness of the reconstruction.

Methods based on neural implicit representations, such as NeRFiller [28] and related work, use generative 2D inpainting to complete missing regions within neural radiance field representations. While these approaches can produce visually plausible completions, the resulting representations are not decomposed at the object level and do not support object-level manipulation or explicit geometric editing.

Gen3DSR [1] reconstructs a dynamic 3D scene from a single RGB image by separating objects from the scene using 2D segmentation masks and completing each independently using depth estimation and generative models. This is conceptually the closest existing approach to the pipeline proposed here, but it relies on a single image as input, limiting geometric accuracy, and does not address the alignment, updating or part-aware scaling stages that are central to this work.

More broadly, the field of semantic scene completion [23] aims to predict complete geometry and semantics from partial observations, but typically treats the scene as a monolithic volume and does not produce object-level decompositions suitable for interactive manipulation.

In summary, while components of the proposed pipeline exist in the literature, no existing work integrates alignment, object-level completion, scene completion and part-aware dynamification into a unified framework operating on multi-source, multi-temporal scan data. This gap motivates the contributions of the subsequent chapters.

2.6 Scope and Assumptions

Having established the terminological and conceptual foundation, we conclude this chapter by summarising the scope and assumptions that govern the remainder of this thesis.

Scene type. The pipeline is designed for single-room indoor environments containing furniture or industrial objects. This constraint reflects the focus on densely furnished scenes where occlusions are significant but the scene structure remains largely planar. Outdoor environments, multi-room layouts and highly organic or curved architectural elements are outside the scope of this work.

Input modalities. The pipeline operates on coloured point clouds and panoramic or localised images. At least one high-quality baseline scan and one set of co-registered images are assumed to be available. The pipeline is designed to accommodate additional incremental observations from lower-end devices.

Object types. The pipeline targets geometrically distinct, rigid objects that are at least partially observed in the scan data. Deformable objects, transparent or highly reflective surfaces, and objects with no observable geometry are not addressed.

Output representation. The DRM is represented as a textured mesh for the background scene and as coloured SDFs or textured meshes for individual objects, with associated part indices. This representation is compatible with standard rendering and game engine pipelines.

Evaluation. Quantitative evaluation is performed primarily on the synthetic V-Scan dataset introduced in Chapter 3, which provides ground-truth geometry for both the scene and individual objects. This synthetic dataset is a methodological necessity rather than a convenience: because real-world scans cannot provide ground truth for occluded regions — the surfaces that completion methods are specifically required to reconstruct — no real-world benchmark can directly assess completion quality in the regions that matter most. Real-world datasets are therefore used for qualitative evaluation and to assess generalisation beyond the synthetic domain.

These assumptions and constraints are revisited in Chapter 9, where the limitations of the proposed pipeline are discussed alongside directions for future work.

Chapter 3

Data Acquisition

3.1 Introduction

The pipeline introduced in Chapter 2 takes as input heterogeneous, partially scanned 3D data of indoor environments. Before any alignment, completion or dynamification can be performed, this data must be acquired. The quality, coverage and format of the input data directly influence every subsequent stage: a scene with severe occlusions will demand more aggressive completion; a misregistered scan will propagate errors through detection and segmentation; a dataset without ground-truth geometry cannot be used to evaluate completion methods quantitatively.

This chapter addresses the data acquisition stage of the pipeline and is structured around two complementary data sources. The first is real-world scan data, which captures authentic geometric and appearance complexity, including irregular occlusion patterns, sensor noise, and furniture arrangements that reflect actual use. The second is synthetic scan data, which sacrifices some realism in exchange for complete, verifiable ground truth — including geometry for regions that were occluded during the scan, which no real-world dataset can provide.

The scope of this chapter, and of the thesis as a whole, is limited to single-room indoor environments containing furniture or industrial equipment. This focus reflects the target application domain of the proposed pipeline and constrains the types of scenes considered. Both residential and institutional interiors are included, as are laboratory and industrial spaces with atypical object classes.

The chapter is structured as follows. Section 3.2 reviews the landscape of

capture technologies and publicly available real-world datasets, and discusses the inherent limitations of both real and synthetic data. Section 3.3 describes the real-world data captures performed within this work and the characteristics of the publicly available datasets used for evaluation. Section 3.4 presents the synthetic data creation methodology and introduces the V-Scan dataset. Section 3.5 summarises the characteristics and limitations of the acquired data and their implications for the subsequent pipeline stages.

3.2 Related Work: Capture Technologies and Datasets

3.2.1 Capture Technologies

The accurate acquisition of 3D indoor geometry has been a central research challenge in robotics, photogrammetry and computer vision for over two decades. The landscape of capture technologies spans a wide range of accuracy, speed, cost and portability trade-offs, and the choice of device has a direct effect on the type and severity of occlusions in the resulting data.

Terrestrial laser scanners. TLS devices are static, tripod-mounted instruments that emit laser pulses in a spherical pattern from a fixed position, recording the distance and direction of each return to produce a dense, highly accurate point cloud. High-end systems such as the Leica ScanStation P30 achieve point-level accuracies below 2 mm at ranges up to 270 m, with scan rates of up to 1 million points per second [3]. Their principal limitation in indoor settings is occlusion: because each setup covers only the hemisphere visible from a single position, any object that is not in the scanner's line of sight from at least one setup will be missing from the data. In practice, acquiring full-room coverage requires between three and ten setups per room, with each additional setup adding scanning and registration time. Furniture, equipment and the room's own structural elements all contribute to residual occlusions even with multiple setups. The resulting data is georeferenced through the use of control targets measured by total station or GNSS, and successive setups are aligned by iterative closest point or target-based registration.

Mobile mapping systems. MMS combine laser scanning with SLAM and IMU data to enable continuous scanning along a trajectory, producing a large-area point cloud from a single walk-through without the need for static setups.

Systems such as the NavVis VLX achieve point accuracies of around 6 mm over 500 m^2, at scan rates of 1.28 million points per second [**?**]. The ability to scan from many positions along a continuous path substantially reduces inter-object occlusions compared to static TLS setups, as the scanner naturally approaches objects from multiple angles. However, trajectory drift and the lower accuracy of SLAM-based localisation compared to surveying-grade control networks introduce systematic errors at larger scales. Indoor MMS devices also typically produce panoramic imagery co-registered to the trajectory, providing appearance data alongside the geometry.

XR and consumer devices. Extended reality devices and consumer smartphones equipped with depth sensors represent the lower end of the accuracy spectrum. The Microsoft Hololens 2 integrates a ToF depth sensor, an 8 MP colour camera, and an on-board SLAM processor, producing a real-time polygonal mesh of the environment at a default spatial resolution of 0.08 m^3 per cell [26]. Consumer smartphones such as the iPhone Pro series embed shorter-range ToF LiDAR sensors with a practical range of around 5 m and an accuracy on the order of 1–2 cm [**?**]. These devices trade geometric precision for speed, portability and cost, enabling rapid partial coverage of a scene in a matter of minutes. Their lower accuracy and limited range make them unsuitable as primary acquisition devices for detailed reconstruction, but they are well suited for incremental updates of existing high-quality datasets, as exploited in the alignment methodology of Chapter 4.

Photogrammetry. SfM photogrammetry reconstructs 3D geometry from a set of overlapping 2D images by jointly estimating camera poses and a sparse 3D point cloud, which can be densified and meshed in post-processing. While photogrammetry is highly flexible and requires no specialised hardware beyond a camera, it is sensitive to textureless or reflective surfaces, requires dense image overlap, and produces lower geometric accuracy than dedicated LiDAR systems without carefully placed control points. In the context of this thesis, photogrammetry is primarily used as a source of localised imagery for alignment and texture reference, rather than as the primary geometry acquisition modality.

3.2.2 Limitations of Real and Synthetic Data

Both real and synthetic data sources present characteristic limitations that are relevant to the proposed pipeline.

Real-world datasets offer authentic geometric and appearance complexity, but they cannot provide ground truth for occluded regions. A scanned point cloud records only what was visible to the sensor; the surfaces behind furniture, under objects and inside cavities are structurally absent from the data. This means that any evaluation of a completion method using only real-world data is limited to assessing the visual plausibility of the result, not its geometric accuracy. Furthermore, real-world datasets require significant capture effort, and the resulting data is heterogeneous in terms of device, acquisition conditions and annotation quality across different scenes.

Synthetic datasets offer complete control over scene geometry, object placement, occlusion configuration and sensor simulation. Because every surface in the virtual scene is known, it is possible to generate both a partial scan mimicking real-world sensor behaviour and a complete ground-truth representation of the scene and its objects. This enables quantitative evaluation of completion methods in terms of geometric accuracy, not merely visual quality. The principal limitation of synthetic data is the domain gap: virtual scenes differ from real environments in lighting, material appearance, geometry complexity and object diversity, which can reduce the transferability of learned models trained exclusively on synthetic data [14].

The approach adopted in this thesis uses both sources in a complementary way. Real-world data is used to validate qualitative performance and to ensure that the pipeline generalises beyond the synthetic domain. Synthetic data, specifically the V-Scan dataset introduced in Section 3.4, provides the quantitative ground truth required for systematic evaluation of each pipeline stage.

3.3 Real-World Data Capture

3.3.1 Laser Scanner Captures

Static TLS captures. A series of real-world indoor scans were acquired at the KU Leuven Technology Campus in Ghent as part of the data collection effort underlying the alignment work in Chapter 4. Three distinct spaces were captured over multiple time periods using a Leica ScanStation P30 and a Leica BLK360. The first space is a soil technology laboratory that resembles an industrial site, housing working desks, machinery and bulk materials. Between periodic data captures spanning several months, the lab remained operational, meaning that all movable objects were displaced between sessions. The second space is a prefab living laboratory, captured from early stake-out through to MEP installation. The third is a residential house in Brugge captured during

refurbishment, with both smartphone and Hololens 2 data supplementing the TLS baseline [26].

For each static TLS capture, between 2 and 20 individual scanner setups were acquired per session, with each setup producing a structured point cloud in E57 format and a set of co-registered panoramic images. Control points established by total station were used to georeference the data. The resulting point clouds span tens to hundreds of millions of points per session, with single-point accuracies below 5 mm. Table 3.1 summarises the captured sessions across all three test cases.

Table 3.1: Overview of the real-world captures used in this thesis, across three test cases and multiple sensing modalities.

Site	**Sensor**	**Sessions**	**Points (M)**	**Images**
Soil lab	Leica P30 / BLK	3	162	241
	Hololens 2	6	1.1	35
Living lab	Leica P30	1	7.6	274
	Photogrammetry	3	1.0	642
	Hololens 2	5	0.4	30
House Brugge	Leica P30	1	265	–
	Photogrammetry	1	–	430
	Hololens 2	20	0.4	91

Static TLS captures provide the highest geometric accuracy of any modality used in this work, but their occlusion pattern is also the most severe. From a limited number of fixed positions, large portions of each room — particularly the rear faces of furniture and the regions beneath and between objects — remain unobserved. This is the primary source of incomplete geometry in the pipeline inputs and motivates the completion methodology developed in Chapters 6 and 7.

Mobile mapping captures. The living laboratory test case was additionally captured using a NavVis VLX indoor MMS. The MMS produced a full-room trajectory scan in a single walk-through session, yielding 162 million points with panoramic images at key trajectory positions. The higher coverage and reduced inter-object occlusion of the MMS data, compared to the static TLS setups, make it a useful reference for assessing the impact of scanner configuration on occlusion severity.

3.3.2 XR Device Captures

The Hololens 2 was used across all three test cases to provide fast, incremental coverage updates to the existing TLS baseline datasets. The device produces a real-time polygonal mesh and a series of localised images captured during the scan session. Scan sessions are short — typically five to fifteen minutes — and can be performed without the logistical overhead of a full TLS campaign. Each session produces between 15 and 500 thousand points, depending on the room size and scan duration, at a spatial resolution of 0.08 m^3 and a point accuracy consistent with the ToF sensor (approximately 10 mm per 10 m of range). The panoramic and localised image streams associated with each session provide appearance data for texture assignment and 2D-based alignment.

A custom Unity-based XR capture application was developed to manage the Hololens 2 data acquisition, enabling real-time feedback on scan coverage and on-device previewing of the accumulated mesh. This application transmits captured meshes and images to a remote server for automatic alignment and integration into the reference dataset, as described in Chapter 4. Android smartphones were used in parallel with the Hololens 2 in some sessions to provide additional image coverage for photogrammetric alignment, exploiting their higher image resolution (up to 12 MP) compared to the Hololens 2 camera (approximately 8 MP effective for XR use).

The principal limitation of XR captures for the purposes of this pipeline is range: the Hololens 2 ToF sensor saturates at distances beyond approximately 3–4 m in practice, meaning that large rooms cannot be scanned completely from a single position and that distant surfaces such as high ceilings or far walls are frequently missing from the data. This limitation makes XR data unsuitable as the primary acquisition modality for completion but appropriate for targeted incremental updates.

3.3.3 Public Real-World Datasets

In addition to the data captured within this work, several publicly available real-world datasets are used for evaluation and comparison throughout the thesis. Table 3.2 provides an overview of the key datasets and their properties.

SUN RGB-D [?]. The SUN RGB-D dataset provides 10,335 RGB-D images captured with four different sensors — Microsoft Kinect v1, Kinect v2, Intel RealSense, and Asus Xtion — across a wide variety of indoor scene types including offices, bedrooms, kitchens and classrooms. The dataset is densely

Table 3.2: Overview of public real-world indoor datasets used in this thesis. GT = ground truth availability for occluded regions.

Dataset	Sensor	Modality	Scenes	
SUN RGB-D [?]	Kinect, Xtion, others	RGB-D images	–	
ScanNet [?]	Structure Sensor	RGB-D video	1,513	
ScanNet++ [?]	FARO laser + DSLR + iPhone	Laser + RGB-D	460	280K
Matterport3D [6]	Matterport Pro Camera	RGB-D panoramas	90	
Redwood-3DScan [?]	PrimeSense Carmine	RGB-D video	–	

annotated and includes 146,617 2D polygons and 64,595 3D bounding boxes with accurate object orientations, as well as a 3D room layout and scene category for each image. SUN RGB-D is widely used as a benchmark for 3D object detection and scene understanding, and it is used in this thesis primarily as a source of evaluation scenes for object detection methods in Chapter 5. Its use of consumer RGB-D sensors means that geometric accuracy is limited, with typical depth noise on the order of several centimetres, and occlusion patterns reflect the single-viewpoint nature of each captured frame.

ScanNet [?]. ScanNet is an RGB-D video dataset containing 2.5 million views in more than 1,500 scans, annotated with 3D camera poses, surface reconstructions, and instance-level semantic segmentations. Scenes were captured with a consumer Structure Sensor mounted on an iPad, producing reconstructed meshes via BundleFusion. The availability of textured meshes and instance-level annotations makes ScanNet a valuable resource for training and evaluating segmentation and completion methods. However, the consumer-grade sensor introduces significant depth noise and limited range, and the handheld capture trajectory results in irregular coverage with many inter-object occlusions unresolved. In this thesis ScanNet is used as a qualitative evaluation dataset for object completion and scene segmentation methods.

ScanNet++ [?]. ScanNet++ presents a large-scale dataset coupling high-quality and commodity-level geometry of indoor scenes. Each scene is captured with a high-end laser scanner at sub-millimetre resolution, along with registered 33-megapixel images from a DSLR camera, and RGB-D streams from an iPhone. Currently, ScanNet++ contains 460 scenes, 280,000 captured DSLR images, and over 3.7M iPhone RGBD frames. The combination of laser-quality geometry and high-resolution imagery makes ScanNet++ the highest-fidelity real-world indoor dataset currently available. The semantic instance annotations further enable evaluation of detection and segmentation methods. In this thesis, ScanNet++ scenes and their pre-labelled instance annotations are used to evaluate the

texture completion methodology of Chapter 6, as reported in the semantic UV mapping work [24].

Matterport3D [6]. Matterport3D is a large-scale RGB-D dataset containing 10,800 panoramic views from 194,400 RGB-D images of 90 building-scale scenes, with annotations including surface reconstructions, camera poses, and 2D and 3D semantic segmentations. Scenes span a wide variety of building types including homes, offices and churches, and each building is captured comprehensively across all floors. The Matterport Pro Camera captures RGB-D panoramas from static positions, producing textured meshes with good overall coverage but the occlusion profile of a limited-position static scanner. Matterport3D is used extensively in this thesis for evaluation of object detection, object completion, scene completion and dynamification methods across Chapters 5 through 8, and its pre-labelled instance masks serve as ground truth for object isolation experiments.

Redwood-3DScan [?]. The Redwood-3DScan dataset contains 10,933 RGB-D scans of individual objects captured by 70 non-expert operators using consumer-grade PrimeSense Carmine cameras, with 9,131 scans categorised into 320 object categories. Unlike the scene-level datasets above, Redwood focuses on individual isolated objects rather than full rooms. Each scan is a short RGB-D video sequence of an operator walking around a single object, from which a partial point cloud and optionally a reconstructed mesh are derived. This makes Redwood valuable as a source of real-world partial object scans for evaluating object completion methods, though the absence of per-object ground truth for occluded regions limits quantitative evaluation to observable surfaces only.

3.3.4 Evaluation and Shortcomings of Real-World Data

Across all real-world datasets and captures, a consistent set of limitations affects their utility for the specific task of evaluating a completion pipeline.

The most fundamental limitation is the absence of ground truth for occluded regions. No real-world scanning campaign, however thorough, can observe the surfaces that are hidden behind or between objects. This means that evaluation of completion quality on real-world data is necessarily qualitative, assessing visual plausibility and consistency rather than geometric accuracy. Evaluating whether a completed surface is geometrically correct — rather than merely visually plausible — requires a ground-truth reference that real-world data cannot provide.

A related limitation concerns object-level ground truth. Even datasets with instance-level annotations, such as Matterport3D and ScanNet, do not provide complete 3D models of individual objects. The annotated instance masks label which points or mesh faces belong to each object, but the hidden surfaces of those objects — the parts not visible from any scan position — are absent. This makes it impossible to evaluate object completion accuracy in terms of the full 3D shape, as opposed to only the visible portion.

Additionally, real-world datasets vary considerably in scan quality, coverage and annotation consistency. Consumer RGB-D sensors introduce geometric noise that can confound completion and segmentation methods. Coverage varies with the scanner configuration: a single-position TLS scan of a cluttered room will have far more severe occlusions than a multi-position scan of the same room. These variations make it difficult to isolate the effect of individual pipeline components across different evaluation scenes.

These shortcomings collectively motivate the synthetic data approach described in the following section.

3.4 Synthetic Data Creation

3.4.1 Related Work: Synthetic Datasets and Virtual Scanners

The use of synthetic data for training and evaluating 3D scene understanding methods has a long history. Scene-level synthetic datasets including SUNCG [**?**], SceneNet RGB-D [**?**] and InteriorNet [**?**] provide large collections of procedurally generated indoor scenes with RGB-D images and dense annotations. At the object level, ShapeNet [7] and ModelNet [**?**] provide thousands of clean 3D CAD models that have been widely used to train completion and classification networks. These datasets have driven substantial progress in 3D learning, but they share a common approach to generating partial scan data: points are sampled directly from mesh surfaces, either uniformly or from simulated viewpoints, without modelling the physical behaviour of the scanning sensor. This produces point distributions that differ systematically from those of real scanners, which measure only line-of-sight surfaces and introduce distance-dependent noise and resolution variation.

A smaller body of work has addressed physically-based virtual scanning simulation. Blume and Heimann [**?**] presented an early laser scanner simulator for wheeled robot navigation. HELIOS++ [29] is a ray-tracing-based aerial LiDAR simulator designed for terrain mapping and forestry analysis, providing

accurate sensor modelling but not designed for indoor scene generation or object-level ground truth. Popovas et al. [21] developed a virtual TLS simulator for training and educational purposes in construction contexts. None of these frameworks provides the combination of indoor procedural scene generation, configurable multi-device simulation, per-object ground truth export, and voxel-based occlusion annotation required for the evaluation tasks of this thesis.

3.4.2 Methodology: The Virtual Scanner

The virtual scanning framework is implemented in Unity and designed to simulate the behaviour of real-world scanning devices through configurable physical parameters [**?**]. Rather than sampling points directly from mesh surfaces, the system performs ray-based scanning from a virtual viewpoint, naturally replicating the line-of-sight visibility constraints of a physical scanner. The full pipeline is summarised in Figure 3.1.

Figure 3.1: Overview of the virtual scanning pipeline showing the scanner setup, ray-based acquisition, panoramic colouring, object isolation and occlusion computation stages.

Scanner setup and parameterisation. The virtual scanner is parameterised by scan density (mm per 10 m), maximum range (m), vertical field of view (degrees), system error (mm) and distance-dependent error (%). Presets are provided that emulate four representative real-world devices: the Leica ScanStation P30 ($\delta_r = 1.2$ mm + 10 ppm, 1 MHz, up to 270 m), the NavVis VLX (6 mm per 500 m^2, 1.28 MHz, up to 50 m), the Leica BLK360 (4 mm at 10 m, 0.68 MHz, up to 45 m), and the iPhone Pro LiDAR (1–2 cm, depth sensing only, approximately 5 m range). This parameterisation allows the same scene to be scanned under multiple simulated device configurations, enabling analysis of the effect of sensor quality on downstream pipeline performance.

Scan vector generation. Scanning begins by generating a spherical array of ray directions. Rays are distributed uniformly along horizontal disks stacked at equal angular intervals in the vertical direction, starting from a configurable minimum elevation angle to model the blind zone beneath the sensor. Equal angular spacing in both directions ensures uniform angular coverage and computational efficiency compared to sequential vertical sweep patterns. The total number of rays is determined by the configured scan density and the device field of view.

Ray-casting and point cloud acquisition. Each ray is cast from the scanner origin until it intersects the collider mesh of a scene object. The intersection point, its distance from the origin and the surface normal at the intersection are recorded. Gaussian noise is applied at the magnitude of the system error parameter, and a distance-proportional error component is added to simulate the increasing measurement uncertainty at longer ranges. Points beyond the maximum range are discarded. The result is a partial, noise-affected point cloud that covers only those surfaces visible from the scanner position, accurately modelling the occlusion pattern of a real static scanner.

Panoramic image capture and point colouring. After the geometric scan is complete, a 360° panoramic image is captured at the scanner's position by rendering a cubemap from the virtual camera and converting it to an equirectangular projection. Each scanned point is then assigned a colour by projecting its 3D position into spherical coordinates and sampling the corresponding pixel in the panoramic image. This produces a coloured point cloud that encodes the lighting conditions, material appearance and textures of the virtual environment, closely matching the output of real TLS systems that capture panoramic imagery alongside their geometric data.

Object-level scanning and isolation. Each non-structural object in the scene is assigned a `ScannedObject` component that stores its oriented bounding box (OBB) and ground-truth mesh reference. After the full-scene scan, each object's partial point cloud is extracted by selecting the scanned points whose positions fall within the object's OBB, optionally expanded by a small margin to avoid clipping edge points. The object's OBB is exported as a set of eight 3D corner points in world coordinates. The ground-truth mesh is simultaneously exported from the scene, providing a complete surface representation of the object independent of the scan coverage.

Voxel-based occlusion computation. Occlusion is computed at both scene level and per-object level using a voxel-based ray-casting approach. For each target — either the full scene bounding box or an individual object's OBB — a regular voxel grid is defined at a configurable resolution. For each voxel centre, a ray is cast from the scanner origin. If the ray intersects any scene geometry before reaching the voxel centre, the voxel is marked as occluded; otherwise it is marked as free. Voxels within the object's surface are marked as occupied. The resulting sparse voxel grids, exported in a compact format, encode the ground-truth visibility information used by the occlusion analysis

stage in Chapter 5 to distinguish sensor occlusion, inter-object occlusion, and contact occlusion as defined in Chapter 2.

Empty scene rescan and export. To generate complete ground-truth geometry for the background scene, the scanner is re-run after removing all non-structural objects from the scene. This produces a complete empty-scene point cloud and panoramic image, corresponding to what the scanner would observe in the absence of furniture. The combination of the furnished scan, the empty-scene scan, the per-object partial clouds, the object ground-truth meshes, and the per-object and full-scene occlusion grids provides all the supervision signals required to evaluate each stage of the pipeline quantitatively.

3.4.3 Procedural Scene Generation

To enable large-scale dataset generation without manual scene authoring, the virtual scanner is integrated with a procedural indoor scene generation system implemented in Unity. The generator produces fully furnished single-room scenes automatically, enabling the creation of a large number of diverse layouts for dataset generation.

Room layouts are generated by sampling room dimensions from a configurable range, producing rectangular floor plans of variable size. The floor is discretised into a regular grid of tiles. Walls are placed along the outer boundary, with each wall segment probabilistically assigned as a solid wall, a window, or a doorway based on predefined probabilities; at least one doorway is always included. Ceiling elements are placed at a fixed height above each floor tile. Furniture is placed in two stages: wall-mounted elements such as shelves and cabinets are first positioned along interior walls with a configurable probability, followed by free-standing items placed within the interior. Collision-free placement is enforced by bounding box intersection tests, producing physically plausible layouts.

The procedural system uses configurable asset collections implemented as Unity ScriptableObjects, each defining a visual style for a room through references to prefab models for structural elements and furniture. Switching between asset configurations generates visually diverse scenes while using the same underlying generation algorithm. All random functions are seeded from a single configurable seed, making generation deterministic and reproducible.

After room generation, the virtual scanner is positioned at the centre of the room bounding box and configured to capture the full scene under a chosen device

preset. The scan is then executed automatically, with no manual intervention required.

3.4.4 The V-Scan Dataset

Using the virtual scanning framework and procedural generation system described above, we introduce the *V-Scan* dataset: a synthetic dataset of partially scanned indoor scenes paired with complete ground-truth geometry at both the scene and object level [**?**].

Dataset composition. The dataset consists of approximately 100 procedurally generated furnished single-room scenes, each scanned using the Leica P30 preset to simulate high-quality TLS acquisition. Each scene in the dataset includes the following data:

- A furnished scene point cloud with colours and surface normals, representing the partial observation from the scanner position.
- An empty scene point cloud with colours and normals, representing the ground-truth background geometry without furniture.
- Equirectangular panoramic images of both the furnished and empty scenes captured at the scanner position.
- A full-scene voxel occlusion grid encoding sensor, inter-object and contact occlusions.
- For each non-structural object: a partial point cloud, the object's oriented bounding box (8 corner points in world coordinates), a per-object voxel occlusion grid, and the ground-truth mesh.

Dataset statistics. Table 3.3 summarises the key statistics of the V-Scan dataset. Scenes contain between 5 and 20 non-structural objects, reflecting the density range of furnished single rooms. The number of points per furnished scene scan ranges from approximately 2 to 8 million, depending on room size and furniture density. Object occlusion levels, measured as the fraction of the object's OBB voxels that are marked as occluded, range from under 10% for small, freestanding objects to over 70% for objects placed against walls or in corners, reflecting the three occlusion categories defined in Chapter 2.

Table 3.3: Summary statistics of the V-Scan dataset.

Property	**Value**
Total scenes	≈ 100
Objects per scene	5 – 20
Points per scene (M)	2 – 8
Mean object occlusion	$\approx 35\%$
Occlusion range	8% – 72%
Voxel grid resolution	0.05 m
Simulated device	Leica P30 preset
Ground truth per object	Full mesh + OBB
Ground truth per scene	Empty scan + pano

Comparison with existing synthetic datasets. Table 3.4 compares V-Scan with the most relevant existing virtual scanning frameworks and synthetic datasets. The key differentiator of V-Scan is the combination of physically plausible ray-based scanning simulation, per-object ground-truth mesh export, and explicit voxel-based occlusion annotation. No existing public dataset provides all three simultaneously for indoor scenes, making V-Scan uniquely suited to the evaluation requirements of the proposed pipeline.

Table 3.4: Comparison of synthetic indoor datasets and virtual scanning frameworks. Ray-based = physically-based ray-casting simulation rather than surface sampling. Per-obj GT = per-object ground-truth mesh. Occ. grid = voxel occlusion annotations.

Dataset / Framework	**Indoor**	**Ray-based**	**Per-obj GT**	**Occ. grid**
SUNCG [**?**]	✓	–	–	–
SceneNet RGB-D [**?**]	✓	–	–	–
InteriorNet [**?**]	✓	–	–	–
Blume et al. [**?**]	✓	✓	–	–
HELIOS++ [29]	Limited	✓	–	–
Popovas et al. [21]	✓	✓	–	–
V-Scan (ours)	✓	✓	✓	✓

3.5 Conclusion

This chapter has described the two data sources used throughout the thesis and characterised their respective properties and limitations. Real-world data, captured across three test sites using static TLS, mobile mapping and XR devices, provides authentic geometric and appearance complexity and enables qualitative validation of pipeline performance. The publicly available datasets

— SUN RGB-D, ScanNet, ScanNet++, Matterport3D and Redwood-3DScan — extend the real-world evaluation scope and provide instance-level annotations for training and benchmarking.

All real-world data sources share a fundamental limitation: they cannot provide ground truth for occluded regions, restricting evaluation to the visible surfaces. This limitation motivates the V-Scan synthetic dataset, which provides complete geometric ground truth at both the scene and object level through physically-based ray-casting simulation, procedural scene generation and explicit occlusion annotation. The combination of both data types allows the subsequent chapters to evaluate completion and segmentation methods both qualitatively, on diverse real-world scenes, and quantitatively, against the V-Scan ground truth.

Table 3.5 provides a final overview of all data sources and their roles within the pipeline evaluation.

Table 3.5: Summary of data sources and their roles in pipeline evaluation.

Dataset	**Role**	**Chapters us**
KU Leuven TLS / MMS / XR captures	Alignment evaluation	4
Matterport3D	Qualitative evaluation, obj. detection	5–8
ScanNet	Qualitative evaluation, segmentation	5, 7
ScanNet++	Texture completion evaluation	6
SUN RGB-D	Object detection evaluation	5
Redwood-3DScan	Object completion evaluation	7
V-Scan (ours)	Quantitative ground-truth evaluation	5–8

Chapter 4

Alignment And Updating

Chapter 5

Scene Segmentation

5.1 Introduction

5.2 Related work

5.3 Methodology

5.3.1 Object detection

5.3.2 Environment segmentation

5.4 Experiments

5.5 Conclusion

Chapter 6

Scene Completion

6.1 Introduction

6.2 Related work

6.3 Methodology

6.3.1 Geometry reconstruction

6.3.2 Color inpainting

6.4 Experiments

6.5 Conclusion

Chapter 7

Object Completion

7.1 Introduction

7.2 Related work

7.3 Methodology

7.3.1 Geometry generation

7.3.2 Texture generation

7.4 Experiments

7.5 Conclusion

Chapter 8

Scene Dynamification

8.1 Introduction

8.2 Related work

8.3 Object Dynamification

8.3.1 Part segmentation

8.3.2 Scaling zone definition

8.3.3 Value interpolation

8.3.4 Value repetition

8.4 Scene Reconstruction

8.5 Experiments

8.6 Conclusion

Chapter 9

Conclusion and future works

9.1 Conclusion

9.2 Future works

Chapter 10

Valorisation

10.1 Introduction

10.2 Proof-of-concept applications

In this section, two tools developed up to the proof-of-concept stage are presented.

10.2.1 Digital upcycling

10.2.2 Furniture Removal

10.3 Valorization potential and market position

10.3.1 Time savings analysis

10.4 Conclusion

Bibliography

[1] Ardelean, A., Özer, M., and Egger, B. Gen3DSR: Generalizable 3D Scene Reconstruction via Divide and Conquer from a Single View, Mar. 2025. arXiv:2404.03421 [cs].

[2] Bassier, M., Bonduel, M., Derdaele, J., and Vergauwen, M. Processing existing building geometry for reuse as Linked Data. *Automation in Construction 115*, March (2020), 103180.

[3] Bassier, M., Vergauwen, M., and Genechten, B. V. Standalone Terrestrial Laser Scanning for Efficiently Capturing Aec Buildings for As-Built Bim. *ISPRS Annals of Photogrammetry, Remote Sensing and Spatial Information Sciences III-6*, July (June 2016), 49–55.

[4] Bassier, M., Vermandere, J., Geyter, S. D., and Winter, H. D. GEOMAPI: Processing close-range sensing data of construction scenes with semantic web technologies. *Automation in Construction 164* (2024), 105454.

[5] Cai, Y., Chen, X., Zhang, C., Lin, K.-Y., Wang, X., and Li, H. Semantic Scene Completion via Integrating Instances and Scene in-the-Loop.

[6] Chang, A., Dai, A., Funkhouser, T., Halber, M., Niessner, M., Savva, M., Song, S., Zeng, A., and Zhang, Y. Matterport3D: Learning from RGB-D Data in Indoor Environments. *International Conference on 3D Vision (3DV)* (2017).

[7] Chang, A. X., Funkhouser, T., Guibas, L., Hanrahan, P., Huang, Q., Li, Z., Savarese, S., Savva, M., Song, S., Su, H., Xiao, J., Yi, L., and Yu, F. ShapeNet: An Information-Rich 3D Model Repository. Tech. rep., Stanford University — Princeton University — Toyota Technological Institute at Chicago, 2015. Issue: arXiv:1512.03012 [cs.GR].

[8] Chen, H., Huang, J., Mu, T.-J., and Hu, S.-M. CIRCLE: Convolutional Implicit Reconstruction and Completion for Large-scale Indoor Scene.

[9] Cheng, Y.-C., Lee, H.-Y., Tulyakov, S., Schwing, A., and Gui, L. SDFusion: Multimodal 3D Shape Completion, Reconstruction, and Generation. In *CVPR* (Dec. 2023).

[10] Chibane, J., Alldieck, T., and Pons-Moll, G. Implicit Functions in Feature Space for 3D Shape Reconstruction and Completion. In *CVPR* (2020).

[11] Dai, A., and Niessner, M. 3DMV : Joint 3D-Multi-View Prediction for 3D Semantic Scene Segmentation. *Proceedings of the European Conference on Computer Vision (ECCV)* (2018).

[12] Dai, A., Ritchie, D., Bokeloh, M., Reed, S., Sturm, J., and Niebner, M. ScanComplete: Large-Scale Scene Completion and Semantic Segmentation for 3D Scans. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition* (2018), 4578–4587. ISBN: 9781538664209.

[13] Dai, A., Siddiqui, Y., Thies, J., Valentin, J., and Niessner, M. SPSG: Self-Supervised Photometric Scene Generation from RGB-D Scans.

[14] De Geyter, S., Bassier, M., and Vergauwen, M. AUTOMATED TRAINING DATA CREATION FOR SEMANTIC SEGMENTATION OF 3D POINT CLOUDS. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences XLVI-5/W1-2022* (Feb. 2022), 59–67.

[15] Heckbert, P. S. Survey of Texture Mapping. *IEEE Computer Graphics and Applications 6*, 11 (Nov. 1986), 56–67.

[16] Hübner, P., Weinmann, M., and Wursthorn, S. Voxel-Based Indoor Reconstruction From HoloLens Triangle Meshes.

[17] Li, M., Li, M., Xu, L., and Wei, M. Hybrid 3D Reconstruction of Indoor Scenes Integrating Object Recognition. *Remote Sensing 16*, 4 (Feb. 2024), 638.

[18] Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., and Ng, R. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis.

[19] Mittal, P., Cheng, Y.-C., Singh, M., and Tulsiani, S. AutoSDF: Shape Priors for 3D Completion, Reconstruction and Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (Mar. 2022), pp. 306–315.

[20] Nunes, L., Marcuzzi, R., Mersch, B., Behley, J., and Stachniss, C. Scaling Diffusion Models to Real-World 3D LiDAR Scene Completion. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (Seattle, WA, USA, June 2024), IEEE, pp. 14770–14780.

[21] Popovas, D., Chizhova, M., Gorkovchuk, D., Gorkovchuk, J., Hess, M., and Luhmann, T. VIRTUAL TERRESTRIAL LASER SCANNER SIMULATOR IN DIGITAL TWIN ENVIRONMENT. In *Proceedings ARQUEOLÓGICA 2.0 - 9th International Congress & 3rd GEORES - GEOmatics and pREServation* (Apr. 2021), Editorial Universitat Politécnica de Valéncia.

[22] Ren, X., Huang, J., Zeng, X., Museth, K., Fidler, S., and Williams, F. XCube (\mathcal{X}^3): Large-Scale 3D Generative Modeling using Sparse Voxel Hierarchies.

[23] Roldão, L., Charette, R. d., and Verroust-Blondet, A. 3D Semantic Scene Completion: A Survey. *International Journal of Computer Vision 130*, 8 (Aug. 2022), 1978–2005.

[24] Vermandere, J., Bassier, M., Cuypers, S., and Vergauwen, M. Semantic UV Mapping to Improve Texture Inpainting for 3D Scanned Indoor Scenes. In *Computer Graphics and Visual Computing (CGVC)* (2024), D. Hunter and A. Slingsby, Eds., The Eurographics Association.

[25] Vermandere, J., Bassier, M., and Vergauwen, M. Automatic alignment and completion of point cloud environments using XR data. In *Proceedings - SIGGRAPH 2022 Posters* (2022).

[26] Vermandere, J., Bassier, M., and Vergauwen, M. Two-Step Alignment of Mixed Reality Devices to Existing Building Data. *Remote Sensing 14*, 11 (2022).

[27] Vermandere, J., Bassier, M., and Vergauwen, M. Geometry and Texture Completion of Partially Scanned 3D Objects Through Material Segmentation:. In *Proceedings of the 20th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications* (Porto, Portugal, 2025), SCITEPRESS - Science and Technology Publications, pp. 193–202.

[28] Weber, E., Hołyński, A., Jampani, V., Saxena, S., Snavely, N., Kar, A., and Kanazawa, A. NeRFiller: Completing Scenes via Generative 3D Inpainting.

[29] Winiwarter, L., Esmorís Pena, A. M., Weiser, H., Anders, K., Martínez Sánchez, J., Searle, M., and Höfle, B. Virtual laser scanning with HELIOS++: A novel take on ray tracing-based simulation of topographic full-waveform 3D laser scanning. *Remote Sensing of Environment 269* (Feb. 2022), 112772.

Statement on the use of Generative AI

I did use the generative AI assistance tools ... Generative AI was used for generating programming code.
The text, code, and images in this thesis are my own (unless otherwise specified). Generative AI has only been used in accordance with the KU Leuven guidelines and appropriate references have been added. I have reviewed and edited the content as needed and I take full responsibility for the content of the thesis.

Curriculum Vitae

Jelle Vermandere was born on June 26, 1995. In 2020, he obtained his master's degree in Architectural Engineering at UGent, graduating with distinction. His thesis focussed on creating a website and database for interactive viewing of 3D models withing exhibitions.

After his studies, Jelle was a self-employed freelancer, creating interactive architectural renderings and video games for various clients.

In July 2021, he started his research at the Faculty of engineering technology at the KU Leuven under the supervision of Prof. Vergauwen and Prof. Bassier. A year later, he secured an FWO strategic basic research grand, allowing him to focus his main research which centers around innovating the built environment through Reality Modelling. The goal of this research is to create dynamic and complete 3D models of indoor environments captured using 3D scan-devices.

Jelle has a wide research interest, focussing around the combination of real and digital techniques. Outside his research, his hobbies led him to mastering the full real-to-digital-to-real workflow using laser scanning, 3D modeling and 3D printing, he can create almost anything. He documents his projects on his YouTube channel where he created a community of over 90.000 subscribes accumulating over 7 million views.

Jelle also likes to practice a wide range of sports including: running, biking, swimming and climbing. After all the hours behind his computer, the physical exercise is a very welcome distraction.

Publications

Journal Articles

1. Vermandere, J., Bassier, M., & Vergauwen, M. (2022). *From Partial Scans to Editable Scenes: Hybrid Object and Scene Completion for Dynamic 3D Environments.* **Automation in Construction**, 14(11), 2680. DOI: 10.3390/rs14112680

2. Bassier, M., Vermandere, J., De Geyter, S., & De Winter, H. (2024). *GEOMAPI: Processing close-range sensing data of construction scenes with semantic web technologies.* **Automation in Construction**, 164, 105454. DOI: 10.1016/j.autcon.2024.105454

3. Vermandere, J., Bassier, M., & Vergauwen, M. (2022). *Two-Step Alignment of Mixed Reality Devices to Existing Building Data.* **Remote Sensing**, 14(11), 2680. DOI: 10.3390/rs14112680

4. De Geyter, S., Vermandere, J., De Winter, H., Bassier, M., & Vergauwen, M. (2022). *Point Cloud Validation: On the Impact of Laser Scanning Technologies on the Semantic Segmentation for BIM Modeling and Evaluation.* **Remote Sensing**, 14(3), 582. DOI: 10.3390/rs14030582

Conference Papers

1. Vermandere, J., Bassier, M., & Vergauwen, M. (2025). *Guided object completion with interactive voxel editing.* **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences**, X-G-2025, 901–906. DOI: 10.5194/isprs-annals-x-g-2025-901-2025

2. Vermandere, J., Bassier, M., & Vergauwen, M. (2025). *Geometry and Texture Completion of Partially Scanned 3D Objects Through Material Segmentation*. In **Proceedings of the 20th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications (GRAPP)** (pp. 193–202). SciTePress. DOI: 10.5220/0013120000003912

3. Vermandere, J., Bassier, M., Cuypers, S., & Vergauwen, M. (2024). *Semantic UV Mapping to Improve Texture Inpainting for 3D Scanned Indoor Scenes*. In D. Hunter & A. Slingsby (Eds.), **Computer Graphics and Visual Computing (CGVC)**. The Eurographics Association. DOI: 10.2312/cgvc.20241221

4. Vermandere, J., Bassier, M., & Vergauwen, M. (2023). *TEXTURE-BASED SEPARATION TO REFINE BUILDING MESHES*. **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences**, X-1/W1-2023, 479–485. DOI: 10.5194/isprs-annals-x-1-w1-2023-479-2023

5. Vermandere, J., Bassier, M., & Vergauwen, M. (2023). *MEASURE UP: A SERIOUS GAME FOR TOPOGRAPHIC EDUCATION*. **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences**, X-5/W1-2023, 69–74. DOI: 10.5194/isprs-annals-x-5-w1-2023-69-2023

6. Vermandere, J., Bassier, M., & Vergauwen, M. (2022). *Automatic alignment and completion of point cloud environments using XR data*. **Proceedings - SIGGRAPH 2022 Posters**.

7. Bassier, M., Vermandere, J., & De Winter, H. (2022). *LINKED BUILDING DATA FOR CONSTRUCTION SITE MONITORING: A TEST CASE*. **ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences**, 5(2), 159–165.

FACULTY OF ENGINEERING TECHNOLOGY
DEPARTMENT OF CIVIL ENGINEERING
Gebroeders de Smetstraat 1
B-9000 Ghent
jelle.vermandere@kuleuven.be
https://iiw.kuleuven.be/onderzoek/geomatics